

Negative LDO (LT3093): Working Document

Digital Audio Player • Rev 9

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1. Summary

Field	Value
Manufacturer / MPN	Analog Devices, LT3093EDD#TRPBF
Package	12-lead DFN, 3 × 3 mm, exposed pad (pin 13) = IN
Function	Ultralow-noise negative LDO, −3V9 → −3.3 V analog rail

2. Design Requirements

Requirement	Value
Input	−3V9 from the LT3462A inverter (regulated), −3.816 to −3.971 V
Output	−3.32 V
Load	I/V, summing, OPA1622 (V− side)
Maximum audio output	0.6 V RMS (EW300: 119 dB/V worst nozzle, 28 Ω ±15%)
Output current	≈ 27 mA quiescent; ≈ 98 mA peak (200 mA rated)
Current limit	125 mA programmed (R_{ILIM})
UVLO	−3.5 V rising
Power good	−3.0 V (90.4% of V_{OUT}), to MCU GPIO
VIOC	Unused, floated
Constraints	Low noise · Small footprint

3. Pin Configuration

Pin	Name	Connection
1, 2	IN	$-3V9$, C_{IN}
3	EN/UV	R_{EN2} from IN, R_{EN1} to GND
4	PG	R_{PU} to $+3V3$, MCU GPIO
5	PGFB	R_{PG1} / R_{PG2} midpoint
6	ILIM	R_{ILIM} to GND (Kelvin to pin 9)
7	VIOC	Not connected (float)
8	SET	$R_{SET} \parallel C_{SET}$ to GND; C_{SET} cathode to SET (SET is at -3.32 V)
9	GND	GND
10	OUTS	OUT at the load
11, 12	OUT	$-3V3A$, C_{OUT} , R_{PG1}
13	Exposed pad	IN ($-3V9$): own pour and via field, never GND

4. External Components

Ref	Value / Rating	MPN	AL-PN
R_{SET}	33.2 k Ω 0.1% 0402	YAGEO RT0402BRD0733K2L	#1-085
C_{SET}	2.2 μ F $\pm 20\%$ tantalum, 16 V, 0603, polarized; anode to GND	Vishay 298D225X0016M2T	#1-098
C_{IN}	10 μ F 10 V X7R 0805 (6.76 μ F effective at 3.9 V)	Murata GCM21BR71A106KE22K	#1-081
C_{OUT}	10 μ F 10 V X7R 0805 (7.56 μ F effective at 3.3 V)	Murata GCM21BR71A106KE22K	#1-081
R_{EN1}	50 k Ω 0.1% 0603	YAGEO RT0603BRD0750KL	#1-075
R_{EN2}	88.7 k Ω 0.1% 0402	YAGEO RT0402BRD0788K7L	
R_{PG1}	135 k Ω 0.1% 0402 (E192)	YAGEO RT0402BRD07135KL	
R_{PG2}	15 k Ω 0.1% 0402	YAGEO RT0402BRD0715KL	#1-078
R_{ILIM}	15.6 k Ω 0.1% 0402 (E192)	YAGEO RT0402BRD0715K6L	
R_{PU}	10 k Ω 0603	Generic	

5. Equations and Results

Output voltage (range over $I_{SET} = 98$ to 102 μ A)

$$V_{OUT} = -I_{SET} R_{SET} = -100 \mu A \times 33.2 k\Omega = -3.32 V \quad (-3.254 \text{ to } -3.386 V)$$

Source: [LT3093 Datasheet \(Analog Devices\)](#), SET pin.

Undervoltage lockout (I_{EN} negligible for $R_{EN1} < 100 \text{ k}\Omega$)

$$R_{EN2} = R_{EN1} \left(\frac{3.5}{1.26} - 1 \right) = 88.89 \text{ k}\Omega \rightarrow 88.7 \text{ k}\Omega$$

$$V_{IN(UVLO)} = -1.26 \left(1 + \frac{88.7}{50} \right) = -3.495 \text{ V} \quad (-3.33 \text{ to } -3.69 \text{ V})$$

$$V_{IN(UVLO,falling)} = -(1.26 - 0.215) \left(1 + \frac{88.7}{50} \right) = -2.90 \text{ V} \quad (\text{sim} \approx -3.0 \text{ V})$$

Source: [LT3093 Datasheet \(Analog Devices\)](#), EN/UV pin.

Power-good threshold (R_{PG1} is the top resistor)

$$R_{PG1} = R_{PG2} \left(\frac{3.0}{0.3} - 1 \right) = 135 \text{ k}\Omega, \quad V_{OUT(PG)} = -0.3 \left(1 + \frac{135}{15} \right) = -3.000 \text{ V} \quad (90.4\%)$$

Source: [LT3093 Datasheet \(Analog Devices\)](#), PGFB pin.

Current limit (ILIM regulates to -300 mV; 1.95 k Ω -A scale factor)

$$I_{LIM} = \frac{1.95 \text{ k}\Omega \cdot \text{A}}{R_{ILIM}} = \frac{1.95 \text{ k}\Omega \cdot \text{A}}{15.6 \text{ k}\Omega} = 125 \text{ mA}$$

Source: [LT3093 Datasheet \(Analog Devices\)](#), ILIM pin.

Soft-start (fast start-up to -3.0 V, then the $R_{SET}C_{SET}$ tail)

$$t_{99} - t_{PG} = \tau \ln \frac{0.32}{0.033} = 2.27 \tau \approx 75 \text{ ms}/\mu\text{F} \quad (2.2 \mu\text{F} : 166 \text{ ms})$$

Source: [LT3093 Datasheet \(Analog Devices\)](#), fast start-up.

Headroom (maximum dropout 280 mV at 100 mA) and dissipation

$$|V_{IN(MIN)}| - |V_{OUT(MIN)}| - V_{DO(MAX)} = 3.816 - 3.287 - 0.28 = 0.25 \text{ V}$$

$$P_D = 0.098 \times (3.981 - 3.32) + 0.004 \times 3.981 \approx 0.08 \text{ W}, \quad \Delta T_J \approx 0.08 \times 34 \approx 2.8^\circ\text{C}$$

Source: [LT3093 Datasheet \(Analog Devices\)](#), dropout and θ_{JA} .

Load current (0.6 V RMS = 0.85 V peak design maximum, both channels in phase):

Case	Per channel pk	Signal total	+ Quiescent	Regulator pk
Full scale, 23.8 Ω (28 Ω -15%)	35.7 mA	71.3 mA	27 mA	$\approx 98 \text{ mA}$
Full scale, 28 Ω nominal	30.3 mA	60.6 mA	27 mA	$\approx 88 \text{ mA}$
110 dB SPL peak (0.35 V RMS), 23.8 Ω	21.1 mA	42.2 mA	27 mA	$\approx 69 \text{ mA}$

5.1 Simulation Results

Test	Result
Dropout (29 Ω / 16.5 Ω load)	203 mV at $\approx 115 \text{ mA}$; 214 mV at $\approx 201 \text{ mA}$

Test	Result
Noise vs C_{SET} (0.47 / 1 / 2.2 / 4.7 μ F)	1.44 / 0.75 / 0.44 / 0.34 μ V RMS (20 Hz to 20 kHz)
Output impedance	0.26 m Ω at 1 kHz; 5.2 m Ω at 20 kHz; 76 m Ω peak at 282 kHz
Start-up (0 $\rightarrow$ -3.9 V in 20 ms)	Regulates at -3.62 V input; PG at -3.000 V after 3.9 ms; settled 166 ms after PG
Brownout (-3.9 $\rightarrow$ 0 V in 50 ms)	Loses regulation at -3.490 V; PG low at -3.129 V; off at -2.957 V
Load step 27 $\leftrightarrow$ 205 mA, 1 μ s edges	-7.6 / +3.6 mV, $\approx$ 2 cycles at 140 kHz, settles in $\approx$ 20 μ s
Line step -3.9 $\rightarrow$ -3.74 V	61 μ V peak, settles in $\approx$ 20 μ s ($\approx$ 68 dB)
Input ripple 7.7 mV pp at 1.2 MHz	1.67 μ V pp at OUT

6. Important Values

Item	Value
SET pin current	100 μ A (98 to 102 μ A over temperature)
EN/UV threshold	-1.26 V (-1.20 to -1.33 V), 215 mV hysteresis
PGFB threshold	-300 mV (288 to 312 mV), 7 mV hysteresis
Dropout	214 mV sim / 330 mV max at 200 mA
Peak load	$\approx$ 98 mA at 0.6 V RMS maximum (49% of rating)
Current limit	125 mA programmed; internal backup 400 mA typ, 220 mA min
GND pin current	2.35 mA quiescent; $\approx$ 7 mA at 200 mA (in dropout)
PSRR at 1.2 MHz	-77.9 dB (simulated)
Output noise	0.44 μ V RMS at 2.2 μ F, 20 Hz to 20 kHz (simulated; model $\approx$ 15% optimistic)
Firmware delay	Wait $\geq$ 170 ms after PG before enabling the OPA1622
PG at cold start	Reads high until IN passes $\approx$ -1.8 V; ignore PG until +3V3 is up
PG in shutdown	Pulled low: reads not-good (opposite of the LT3042)
C_{OUT} requirement	$\geq$ 4.7 μ F, ESR < 30 m Ω , ESL < 1.5 nH
θ_{JA} (DFN)	34 $^{\circ}$ C/W

7. Open Problems

#	Problem
1	Exposed pad is IN (-3V9): dedicated pour and via field; the footprint must not tie it to GND.
2	DAC analog supply allocation to this rail.
3	Confirm package (DFN vs MSOP-12E) and the R_{EN2} / R_{PG1} part numbers.
4	PG brownout warning is 2.2 ms at the simulated ramp rate; a real brownout is faster. Consider a hardware mute from PG.
5	PG low is specified at 100 μ A; a 10 k Ω pull-up to +3V3 sinks 330 μ A. Verify or raise R_{PU} .
6	ILIM scale-factor tolerance: confirm the minimum limit stays above 98 mA; confirm the R_{ILIM} part number.
7	Remaining simulations: current limit at 125 mA, PG in shutdown, full chain with the LT3462A.

8. Revision Log

Rev	Date	Change
0	2026-09-22	Initial issue: R_{SET} , UVLO, PG, soft-start, headroom, dissipation.
1	2026-09-22	PG divider 135 k Ω / 15 k Ω , threshold -3.0 V.
2 to 4	2026-09-22	Simulation results: dropout, PSRR, noise, output impedance, start-up, brownout, load and line steps.
5	2026-09-22	Maximum output 0.6 V RMS; peak load 98 mA.
6	2026-09-22	Current limit 125 mA: R_{ILIM} 15.6 k Ω .
7	2026-09-23	Input from the LT3462A inverter.
8	2026-09-23	C_{SET} 2.2 μ F tantalum, cathode to SET.
9	2026-09-25	Reformatted; linked contents and sources; quiescent and 2.7 MHz PSRR items removed.

9. Sources

1. [LT3093 Datasheet \(Analog Devices\)](#)